Cell Biology: Study Notes

1. Cell Theory

Cell theory is one of the foundations of biology. It has three main ideas: all living things are made of one or more cells, the cell is the basic unit of structure and function in living things, and all cells come from pre-existing cells.

The first two ideas came from the work of Matthias Schleiden, who studied plants, and Theodor Schwann, who studied animals, in the late 1830s. Rudolf Virchow added the third idea in 1855 with the phrase 'omnis cellula e cellula'.

Cells are usually microscopic. A typical animal cell is between 10 and 30 micrometres across, while most bacteria are only 1 to 5 micrometres long. Their small size keeps the surface area to volume ratio high, which lets nutrients and waste move in and out quickly enough.

2. Prokaryotic and Eukaryotic Cells

Prokaryotic cells, such as bacteria and archaea, do not have a nucleus. Their DNA is a single circular chromosome found in a region called the nucleoid. Many prokaryotes also carry small rings of DNA called plasmids, which often contain genes for antibiotic resistance.

Eukaryotic cells, found in animals, plants, fungi and protists, keep their DNA inside a membrane-bound nucleus. They also contain other membrane-bound organelles that divide the cell into compartments with different jobs.

Both cell types have a cell membrane, cytoplasm and ribosomes. Prokaryotic ribosomes are smaller (70S) than the ribosomes in the eukaryotic cytoplasm (80S), a difference that some antibiotics use to target bacteria.

3. Organelles and Their Functions

The nucleus stores the cell's DNA and controls gene expression. It is surrounded by a double membrane called the nuclear envelope, which has pores that let RNA and proteins pass through. The nucleolus inside the nucleus makes ribosomal RNA.

Mitochondria are often called the powerhouse of the cell because they produce most of its ATP through aerobic respiration. They have their own circular DNA and a folded inner membrane; the folds are called cristae and increase the surface area for respiration.

Ribosomes build proteins by reading messenger RNA. Free ribosomes make proteins used inside the cell, while ribosomes attached to the rough endoplasmic reticulum make proteins that will be exported or placed in membranes.

The smooth endoplasmic reticulum makes lipids and helps detoxify drugs. The Golgi apparatus modifies, sorts and packages proteins into vesicles. Lysosomes contain digestive enzymes that break down worn-out organelles and material taken in by the cell.

Plant cells also have chloroplasts, where photosynthesis happens, a large central vacuole that stores water and keeps the cell firm, and a cell wall made of cellulose.

4. The Cell Membrane and Transport

The cell membrane is described by the fluid mosaic model: a phospholipid bilayer with proteins floating in it. Phospholipids have hydrophilic heads that face the water on both sides and hydrophobic tails that face inward. Cholesterol sits between the phospholipids and helps control fluidity.

Diffusion is the net movement of particles from an area of higher concentration to an area of lower concentration. It is passive, meaning it does not need energy from the cell. Small non-polar molecules such as oxygen and carbon dioxide diffuse straight through the bilayer.

Facilitated diffusion uses channel or carrier proteins to help larger or charged particles cross the membrane, still moving down their concentration gradient. Osmosis is the diffusion of water across a partially permeable membrane from a dilute solution to a more concentrated one.

Active transport moves substances against their concentration gradient and needs energy from ATP. The sodium-potassium pump is a classic example: each cycle pumps three sodium ions out of the cell and two potassium ions in.

5. Cell Division

Mitosis produces two genetically identical daughter cells and is used for growth and repair. Its stages are prophase, metaphase, anaphase and telophase, usually followed by cytokinesis, which splits the cytoplasm.

During prophase the chromosomes condense and the spindle starts to form. In metaphase the chromosomes line up along the middle of the cell. In anaphase the sister chromatids are pulled to opposite poles, and in telophase new nuclear envelopes form around each set of chromosomes.

Meiosis makes gametes. It involves two divisions and produces four cells, each with half the original number of chromosomes. Crossing over and independent assortment during meiosis create genetic variation.

6. Cellular Respiration

Aerobic respiration releases energy from glucose using oxygen. The overall equation is glucose +

oxygen -> carbon dioxide + water, with the energy stored as ATP.

Glycolysis happens in the cytoplasm and splits one glucose molecule into two pyruvate molecules, giving a small net gain of 2 ATP. The link reaction and the Krebs cycle take place in the mitochondrial matrix and release carbon dioxide.

Most ATP is made by oxidative phosphorylation on the inner mitochondrial membrane, where the electron transport chain uses oxygen as the final electron acceptor. In total, one glucose molecule yields roughly 30 to 32 ATP.

Without oxygen, cells rely on anaerobic respiration. Human muscle cells produce lactate, while yeast produces ethanol and carbon dioxide. Anaerobic respiration releases far less ATP than aerobic respiration.